Reduced Optimal Control with PDE Constraint: Model, Discretization, and First-Order Methods

Fachseminar

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Abstract

1 Problem Statement

We consider a linear-quadratic optimal control problem governed by the Poisson equation on a polygonal domain $\Omega \subset \mathbb{R}^2$ (in computations, typically $[0,1]^2$). A circular subregion $\omega \subset \Omega$ is tagged on the mesh and serves as the support of the control. The control acts as a volumetric source term restricted to ω:

$$-\Delta y = \chi_\omega \, u \quad \text{in } \Omega, \tag{1}$$

$$y = 0 \quad \text{on } \partial\Omega, \tag{2}$$

where χ_ω denotes the characteristic function of ω. We denote by $V = H_0^1(\Omega)$ the state space and by $U = L^2(\Omega)$ the control space. For a given target $y_d \in L^2(\Omega)$ and a regularization parameter $\beta > 0$, we minimize

$$J(y,u) = \tfrac{1}{2} \int_\Omega (y - y_d)^2 \, dx + \tfrac{\beta}{2} \int_\Omega u^2 \, dx. \tag{3}$$

The target is chosen as the pyramid-like profile

$$y_d(x) = 0.5 - \max\left(|x_1 - 0.5|,\, |x_2 - 0.5|\right), \tag{4}$$

which belongs to $L^2(\Omega)$. The control enters only on ω via χ_ω, so the effective source equals u on ω and vanishes on $\Omega \setminus \omega$. The PDE with homogeneous Dirichlet boundary conditions is well-posed for $u \in L^2(\Omega)$, yielding a unique weak solution $y \in V$ by Lax–Milgram.

2 Variational Formulation and Discretization

Weak formulation. Define the bilinear and linear forms

$$a(y,v) = \int_\Omega \nabla y \cdot \nabla v \, dx, \qquad b(u,v) = \int_\omega u \, v \, dx. \tag{5}$$

For any $u \in L^2(\Omega)$, the weak state equation reads: find $y \in V$ such that

$$a(y,v) = b(u,v) \quad \forall v \in V. \tag{6}$$

By coercivity of $a(\cdot,\cdot)$ on V and boundedness of $b(\cdot,\cdot)$, this admits a unique $y \in V$.

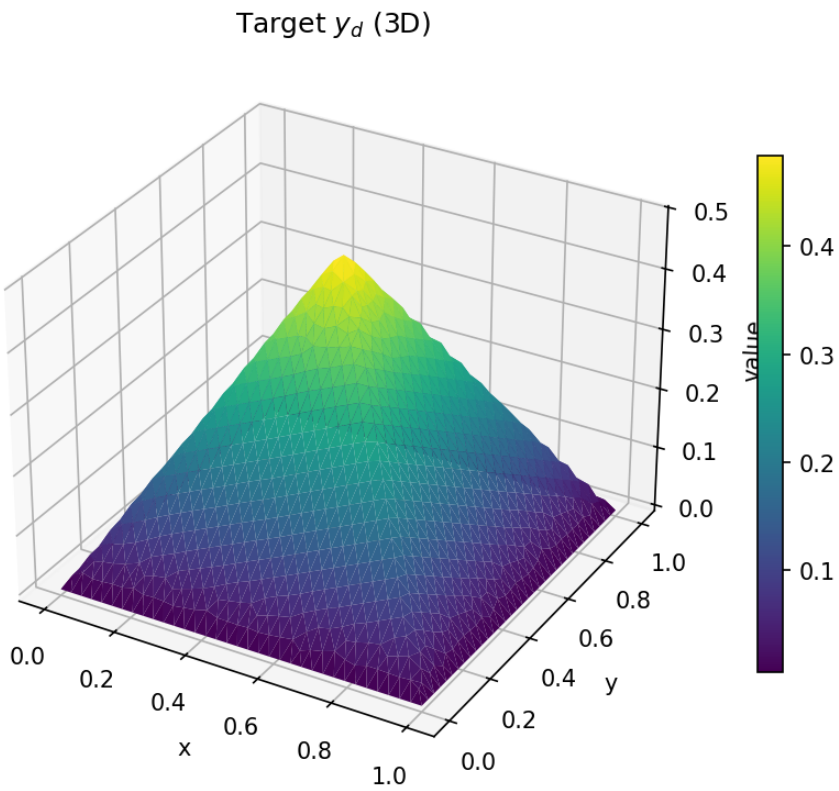

Figure 1: 3D surface of the target profile y_d on Ω.

Finite elements. Let $\mathcal{T}_h$ be a conforming triangulation of Ω. We use the conforming P1 space $V_h \subset V$ with nodal basis $\{\varphi_i\}_{i=1}^n$. The control is represented in the same P1 basis but measured with the L^2 inner product ("control metric"). We assemble the matrices

$$A_{ij} = \int_\Omega \nabla\varphi_j \cdot \nabla\varphi_i \, dx, \quad M_{ij} = \int_\Omega \varphi_j \, \varphi_i \, dx, \quad B_{ij} = \int_\omega \varphi_j \, \varphi_i \, dx, \quad (M_U)_{ij} = \int_\Omega \varphi_j \, \varphi_i \, dx. \tag{7}$$

Here, A is the stiffness matrix for state/adjoint solves, M is the state mass matrix used to weight the tracking term, B maps control coefficients to the state RHS via restriction to ω, and M_U is the control-space mass matrix defining the L^2 inner product in U_h.

Boundary conditions. Homogeneous Dirichlet boundary conditions are imposed on the state/adjoint by modifying A and zeroing the RHS at boundary degrees of freedom when solving. Importantly, no boundary conditions are applied to B or M_U; these operators remain the unmodified L^2-based forms on Ω.

Discrete state equation and objective. With coefficient vectors $\mathbf{y}, \mathbf{u} \in \mathbb{R}^n$ and target coefficients $\mathbf{y}_d \in \mathbb{R}^n$, the discrete state equation and cost read

$$A\,\mathbf{y} = B\,\mathbf{u}, \tag{8}$$

$$J(\mathbf{y}, \mathbf{u}) = \tfrac{1}{2}(\mathbf{y} - \mathbf{y}_d)^\top M(\mathbf{y} - \mathbf{y}_d) + \tfrac{\beta}{2}\mathbf{u}^\top M_U\,\mathbf{u}. \tag{9}$$

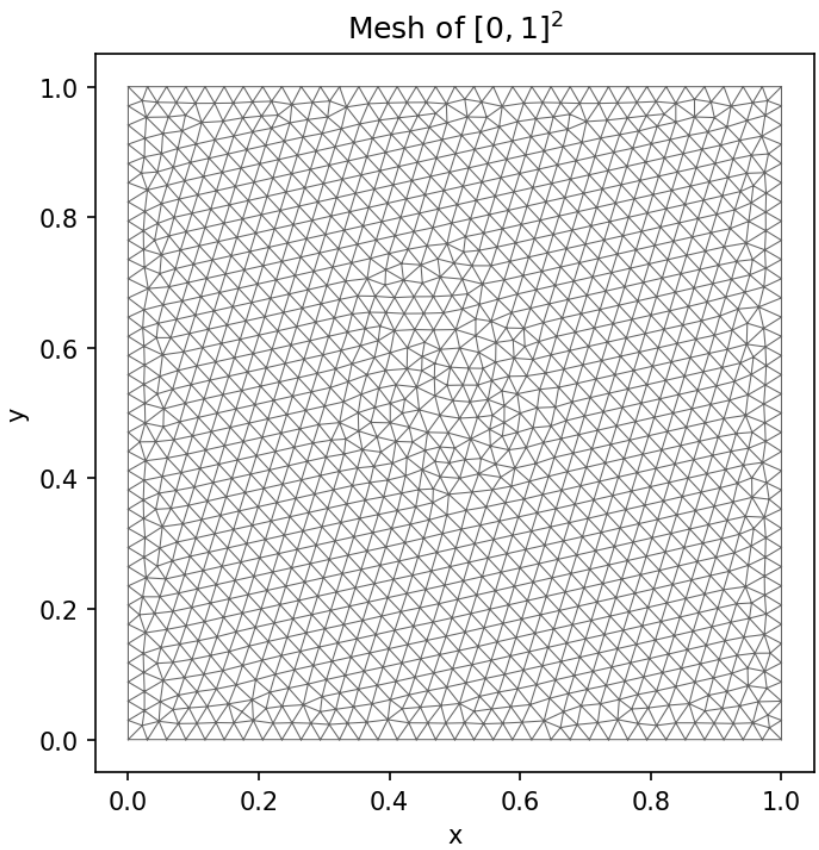

Figure 2: Triangular mesh of Ω used for the P1 finite element discretization.

3 Reduced Formulation

State elimination. Eliminating the state via the discrete PDE constraint gives $\mathbf{y}(\mathbf{u}) = A^{-1}B\,\mathbf{u}$ and the reduced functional $F\colon \mathbb{R}^n \to \mathbb{R}$:

$$F(\mathbf{u}) = \tfrac{1}{2}(A^{-1}B\,\mathbf{u} - \mathbf{y}_d)^\top M(A^{-1}B\,\mathbf{u} - \mathbf{y}_d) + \tfrac{\beta}{2}\,\mathbf{u}^\top M_U\,\mathbf{u}. \tag{10}$$

Adjoint and gradient (continuous view). Introducing an adjoint variable $p \in V$ and the Lagrangian

$$\mathcal{L}(y, u, p) = \tfrac{1}{2}\|y - y_d\|^2_{L^2(\Omega)} + \tfrac{\beta}{2}\|u\|^2_{L^2(\Omega)} + a(y, p) - b(u, p), \tag{11}$$

the optimality system reads: state $a(y, v) = b(u, v)$, adjoint $a(v, p) = (y - y_d, v)$ for all $v \in V$, and variational gradient condition $(\beta u + \chi_\omega p, w)_{L^2(\Omega)} = 0$ for all $w \in U$. Hence, formally, $\beta u + p\,\chi_\omega = 0$ in Ω.

Adjoint and gradient (discrete view). In coefficients, the adjoint solves

$$A\,\mathbf{p} = M\,(\mathbf{y}(\mathbf{u}) - \mathbf{y}_d), \tag{12}$$

and the gradient with respect to the control space inner product induced by M_U is

$$\nabla F(\mathbf{u}) = M_U^{-1}\,B^\top \mathbf{p} + \beta\,\mathbf{u}. \tag{13}$$

The first-order optimality condition $\nabla F(\mathbf{u}_*) = 0$ is equivalent to

$$B^\top \mathbf{p}_* + \beta\,M_U\,\mathbf{u}_* = 0, \qquad A\,\mathbf{p}_* = M\,(\mathbf{y}(\mathbf{u}_*) - \mathbf{y}_d). \tag{14}$$

Hessian, convexity, and Lipschitz constant. The reduced Hessian applied to $\mathbf{v}$ is

$$H\,\mathbf{v} = M_U^{-1}\,B^\top A^{-1} M A^{-1} B\,\mathbf{v} + \beta\,\mathbf{v}. \tag{15}$$

Since A, M, and M_U are symmetric positive definite on their natural subspaces and B is linear, the quadratic form associated with H is strictly positive for $\beta > 0$; thus, F is strictly convex with a unique minimizer. In the U-metric, the gradient is Lipschitz with constant

$$L = \left\| M_U^{-1/2} B^\top A^{-1} M A^{-1} B\, M_U^{-1/2} \right\|_2 + \beta, \tag{16}$$

namely the spectral radius of the SPD term plus β.

4 Optimization Methods

We first rewrite the reduced problem as a quadratic optimization problem in the control coefficients. Using the discrete operators, the objective

$$F(\mathbf{u}) = \tfrac{1}{2}(A^{-1}B\,\mathbf{u} - \mathbf{y}_d)^\top M (A^{-1}B\,\mathbf{u} - \mathbf{y}_d) + \tfrac{\beta}{2}\,\mathbf{u}^\top M_U\,\mathbf{u}$$

expands to the unconstrained QP

$$\min_{\mathbf{u}\in\mathbb{R}^n}\ \tfrac{1}{2}\,\mathbf{u}^\top Q\,\mathbf{u} + \mathbf{c}^\top \mathbf{u} + \text{const}, \qquad Q = B^\top A^{-1} M A^{-1} B + \beta\, M_U, \quad \mathbf{c} = -\,B^\top A^{-1} M\,\mathbf{y}_d, \tag{17}$$

with constant term $\text{const} = \frac{1}{2}\,\mathbf{y}_d^\top M\,\mathbf{y}_d$. Since A, M, and M_U are SPD on their natural subspaces, Q is SPD, so the QP is strictly convex with unique minimizer characterized by the normal equations

$$Q\,\mathbf{u}_* = B^\top A^{-1} M\,\mathbf{y}_d. \tag{18}$$

Equivalence with reduced KKT. Rearranging the normal equations gives

$$\beta\, M_U\,\mathbf{u}_* \;+\; B^\top A^{-1} M\,(A^{-1}B\,\mathbf{u}_* - \mathbf{y}_d) \;=\; 0.$$

Defining the adjoint by $A\,\mathbf{p}_* = M\,(A^{-1}B\,\mathbf{u}_* - \mathbf{y}_d)$ (i.e., $\mathbf{p}_* = A^{-1}M\,(A^{-1}B\,\mathbf{u}_* - \mathbf{y}_d)$) yields the reduced optimality condition

$$B^\top \mathbf{p}_* + \beta\, M_U\,\mathbf{u}_* = 0,$$

which is exactly the KKT stationarity written in Section 3.

We solve the unconstrained, strictly convex reduced problem in the control space endowed with the L^2 inner product induced by M_U:

$$\min_{\mathbf{u}\in\mathbb{R}^n}\ F(\mathbf{u}) = \tfrac{1}{2}(A^{-1}B\,\mathbf{u} - \mathbf{y}_d)^\top M (A^{-1}B\,\mathbf{u} - \mathbf{y}_d) + \tfrac{\beta}{2}\,\mathbf{u}^\top M_U\,\mathbf{u}. \tag{19}$$

The control-space inner product and norm are

$$\langle \mathbf{a}, \mathbf{b}\rangle_U := \mathbf{a}^\top M_U\,\mathbf{b}, \qquad \|\mathbf{a}\|_U := \sqrt{\langle \mathbf{a}, \mathbf{a}\rangle_U}. \tag{20}$$

The gradient used by all methods is the Riesz-represented gradient in U:

$$\nabla F(\mathbf{u}) = M_U^{-1}\,B^\top \mathbf{p} + \beta\,\mathbf{u}, \quad \text{where } A\,\mathbf{p} = M\,(\mathbf{y}(\mathbf{u}) - \mathbf{y}_d),\ A\,\mathbf{y}(\mathbf{u}) = B\,\mathbf{u}. \tag{21}$$

Unless stated otherwise, we initialize at $\mathbf{u}_0 = \nabla F(\mathbf{0})$ (as in the code) and stop when $\|\nabla F(\mathbf{u}_k)\|_U \le$ `tol` or a maximum number of iterations is reached.

Algorithm 1: Gradient Descent in U with step $\alpha = 1/L$

```
Data: tolerance tol > 0, max_iter, model with ∇F, ‖·‖_U
Result: sequence (u_k) converging to a minimizer
u ← u_0;
;                                                    // default ∇F(0)
Estimate L by power iteration; α ← 1/L;
for k = 0, 1, ... do
    g ← ∇F(u); g_n ← ‖g‖_U;
    if g_n ≤ tol then
        break
    end
    u ← u − α g;
end
```

4.1 Gradient Descent with Fixed Step $\alpha = 1/L$

For an L-Lipschitz gradient in the U-metric, the fixed-step update

$$\mathbf{u}_{k+1} = \mathbf{u}_k - \alpha\,\nabla F(\mathbf{u}_k), \quad \alpha = 1/L, \tag{22}$$

produces a monotone decrease in F for convex F. In practice, L is estimated via a matrix-free power iteration on the Hessian action $H\,\mathbf{v} = M_U^{-1}B^\top A^{-1}MA^{-1}B\,\mathbf{v} + \beta\,\mathbf{v}$.

4.2 Barzilai–Borwein (Alternating BB1/BB2)

BB methods approximate inverse curvature along the last step by selecting the steplength from secant information in the U-inner product. Define $\mathbf{s}_k = \mathbf{u}_k - \mathbf{u}_{k-1}$, $\mathbf{d}_k = \nabla F(\mathbf{u}_k) - \nabla F(\mathbf{u}_{k-1})$, and the U-inner product $\langle\cdot,\cdot\rangle_U$. The two classical choices are

$$\alpha_k^{\mathrm{BB1}} = \frac{\langle \mathbf{d}_k, \mathbf{d}_k\rangle_U}{\langle \mathbf{s}_k, \mathbf{d}_k\rangle_U}, \qquad \alpha_k^{\mathrm{BB2}} = \frac{\langle \mathbf{s}_k, \mathbf{s}_k\rangle_U}{\langle \mathbf{s}_k, \mathbf{d}_k\rangle_U}. \tag{23}$$

We alternate BB1/BB2 per iteration and fall back to $\alpha_k = 1$ if $\left|\langle \mathbf{s}_k, \mathbf{d}_k\rangle_U\right|$ is numerically tiny. The update is $\mathbf{u}_{k+1} = \mathbf{u}_k - (1/\alpha_k)\,\nabla F(\mathbf{u}_k)$.

4.3 Nesterov Acceleration (FISTA-Style, Step $1/L$, Optional Restart)

We employ a FISTA-style accelerated scheme in the U-metric with step $\alpha = 1/L$. Let $\mathbf{y}_k$ be the extrapolated point and t_k the momentum parameter. A simple restart resets momentum when the objective increases.

$$\mathbf{u}_{k+1} = \mathbf{y}_k - \alpha\,\nabla F(\mathbf{y}_k), \quad t_{k+1} = \tfrac{1}{2}(1 + \sqrt{1 + 4t_k^2}), \quad \beta_k = \frac{t_k - 1}{t_{k+1}}, \quad \mathbf{y}_{k+1} = \mathbf{u}_{k+1} + \beta_k(\mathbf{u}_{k+1} - \mathbf{u}_k). \tag{24}$$

With restart, if $F(\mathbf{u}_{k+1}) > F(\mathbf{u}_k)$ then set $\mathbf{y}_{k+1} \leftarrow \mathbf{u}_k$, $t_{k+1} \leftarrow 1$.

4.4 Nesterov Acceleration with Known (m, L) (Constant Parameters)

For the strongly convex quadratic QP with Hessian Q in the U-metric, let $L = \lambda_{\max}$ and $m = \lambda_{\min}$ denote the largest/smallest generalized eigenvalues of the pencil (Q, M_U), equivalently the spectrum of $H = M_U^{-1/2}QM_U^{-1/2}$. Set

$$\alpha = 1/L, \qquad \kappa = L/m, \qquad \beta = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}.$$

Algorithm 2: Barzilai–Borwein in U (alternating BB1/BB2)

```
Data: tol > 0, max_iter, model with ∇F, ⟨·,·⟩_U, ‖·‖_U
Result: sequence (u_k)
u_{-1} ← 0; g_{-1} ← ∇F(0);
u_0 ← u_0;
g_0 ← ∇F(u_0);
;                                                    // default g_{-1}
for k = 0, 1, ... do
    if ‖g_k‖_U ≤ tol then
        break
    end
    s_k ← u_k − u_{k-1}; d_k ← g_k − g_{k-1}; sd ← ⟨s_k, d_k⟩_U;
    if |sd| < ε then
        α_k ← 1;
    else
        if k is even then
            α_k ← ⟨d_k, d_k⟩_U / sd;
        else
            α_k ← sd / max{⟨s_k, s_k⟩_U, ε};
        end
    end
    u_{k+1} ← u_k − (1/α_k) g_k;
    u_{k-1} ← u_k; g_{k-1} ← g_k;
    u_k ← u_{k+1}; g_k ← ∇F(u_k);
end
```

Algorithm 3: Nesterov acceleration (FISTA-style, step $1/L$, restart)

```
Data: tol > 0, max_iter, model with ∇F, F; estimate L
Result: sequence (u_k)
α ← 1/L;
u_0 ← u_0;
y_0 ← u_0;
t_0 ← 1;
f_prev ← F(u_0);
;                                                    // default ∇F(0)
for k = 0, 1, ... do
    g ← ∇F(y_k); g_n ← ‖g‖_U;
    if g_n ≤ tol then
        break
    end
    u_{k+1} ← y_k − α g; f_next ← F(u_{k+1});
    if restart and f_next > f_prev then
        y_{k+1} ← u_k; t_{k+1} ← 1; f_prev ← F(u_k);
    else
        t_{k+1} ← (1/2)(1 + sqrt(1 + 4 t_k^2)); β_k ← (t_k − 1)/t_{k+1};
        y_{k+1} ← u_{k+1} + β_k (u_{k+1} − u_k);
        u_k ← u_{k+1}; t_k ← t_{k+1}; f_prev ← f_next;
    end
end
```

Then the constant-parameter Nesterov scheme reads

$$\mathbf{y}_k = \mathbf{u}_k + \beta\,(\mathbf{u}_k - \mathbf{u}_{k-1}), \qquad \mathbf{u}_{k+1} = \mathbf{y}_k - \alpha\,\nabla F(\mathbf{y}_k), \tag{25}$$

and achieves a linear rate $\|\mathbf{u}_k - \mathbf{u}_*\|_U \le C\,\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k$. In practice, we compute (L, m) by solving the generalized eigenproblem (Q, M_U) with $Q\,v = \lambda\,M_U\,v$, where $Qv = B^T A^{-1} M A^{-1} B v + \beta M_U v$ is applied matrix-free via $Qv = M_U\,Hv$.

Algorithm 4: Nesterov with known (m, L) (constant β)

```
Data: tol > 0, max_iter, model with ∇F, M_U, Hessian action H
Result: sequence (u_k)
Estimate (L, m) from generalized eigenpairs of (Q, M_U); α ← 1/L; κ ← L/m;
 β ← (√κ − 1)/(√κ + 1)
u_{−1} ← u_0; u ← u_0;
for k = 0, 1, ... do
    y ← u + β(u − u_{k−1});
    g ← ∇F(y); g_n ← ‖g‖_U;
    if g_n ≤ tol then
        break
    end
    u_{k+1} ← y − α g;
end
```

4.5 Remarks on Metrics and Implementation

All inner products and norms above are in the control metric induced by M_U. This choice ensures that steplengths and convergence tests reflect the natural topology of the control space (consistent with regularization). The adjoint reuse of A facilitates efficient linear solves and preconditioning.

5 Numerical Results

We evaluate the four first-order methods on the external-mesh reduced model with control region tag $\omega = 2$ and Tikhonov parameter $\beta = 10^{-3}$. The initialization is $\mathbf{u}_0 = \nabla F(\mathbf{0})$ and the stopping criterion is $\|\nabla F(\mathbf{u}_k)\|_U \le \texttt{tol}$ or a cap on iterations.

The Lipschitz constant used by fixed-step methods is estimated via power iteration, yielding $L \approx 1.0010 \times 10^{-3}$. For constant-parameter Nesterov, the generalized eigenvalues of the Hessian in the U-metric satisfy $m \approx 1.0000 \times 10^{-3}$ and $L \approx 1.3072 \times 10^{-3}$.

Method	Final cost $F(\mathbf{u}_k)$	Iterations
BB	0.0162550	4
GD $(1/L)$	0.0162550	12
Nesterov $(1/L$ + restart)	0.0162550	102
Nesterov (m, L)	0.0162550	6

Table 1: Final objective value and iteration counts. All methods reach the same optimum; BB attains it in the fewest iterations.

Overall, all methods converge to the same minimizer of the strictly convex reduced problem, corroborating the quadratic structure. In our run, Barzilai–Borwein required only 4 iterations to reach tolerance, while fixed-step GD and constant-parameter Nesterov took 12 and 6 iterations,

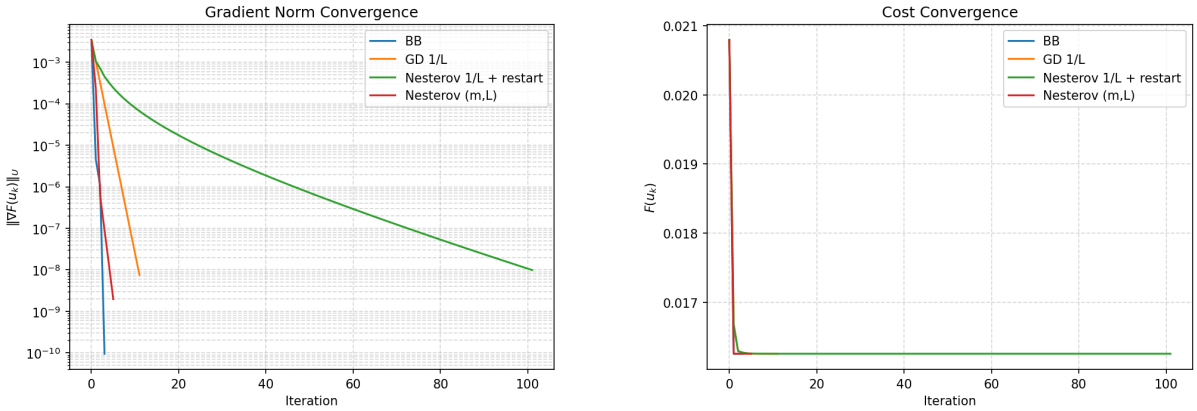

Figure 3: Convergence of gradient norm (left) and objective value (right). The control-space norm $\|\cdot\|_U$ is induced by M_U.

respectively; the FISTA-style Nesterov with restart was significantly slower (102 iterations) under the chosen parameters.

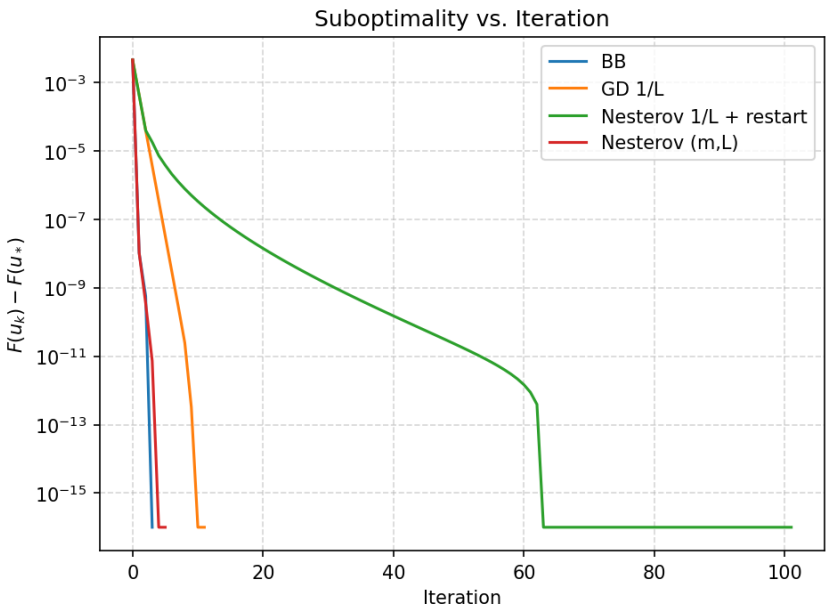

Figure 4: Suboptimality $F(\mathbf{u}_k) - F(\mathbf{u}_*)$ on a logarithmic scale, highlighting linear convergence for strongly convex quadratics and relative speed differences among methods.

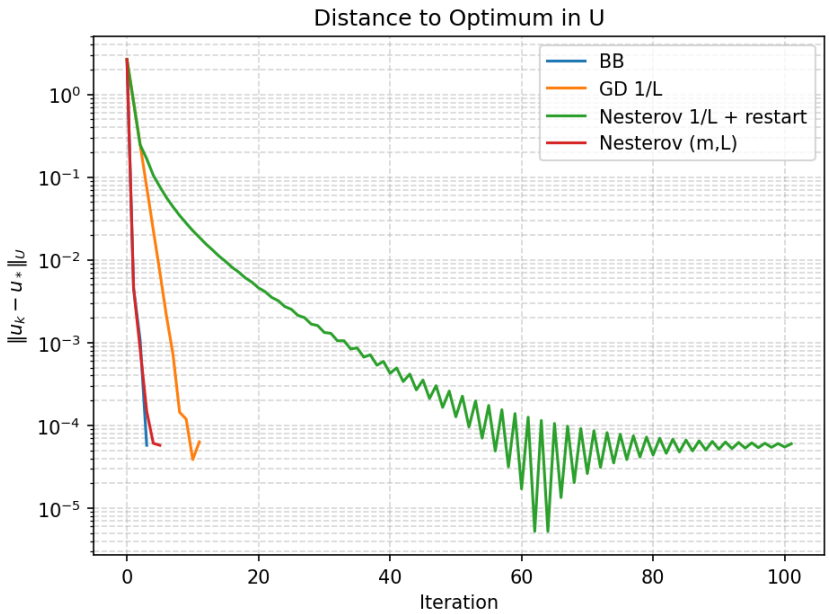

Figure 5: Distance to the optimizer in the control metric, $\|\mathbf{u}_k - \mathbf{u}_*\|_U$, confirming convergence of all methods to the unique minimizer and illustrating rate differences.

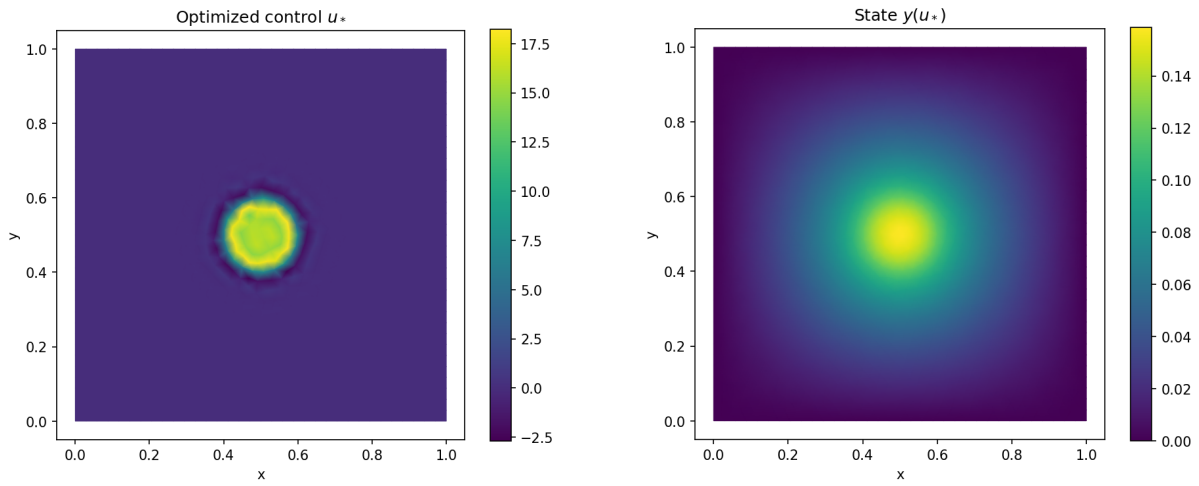

Figure 6: Optimized control u_* (left) supported on ω and the corresponding state $y(u_*)$ (right).